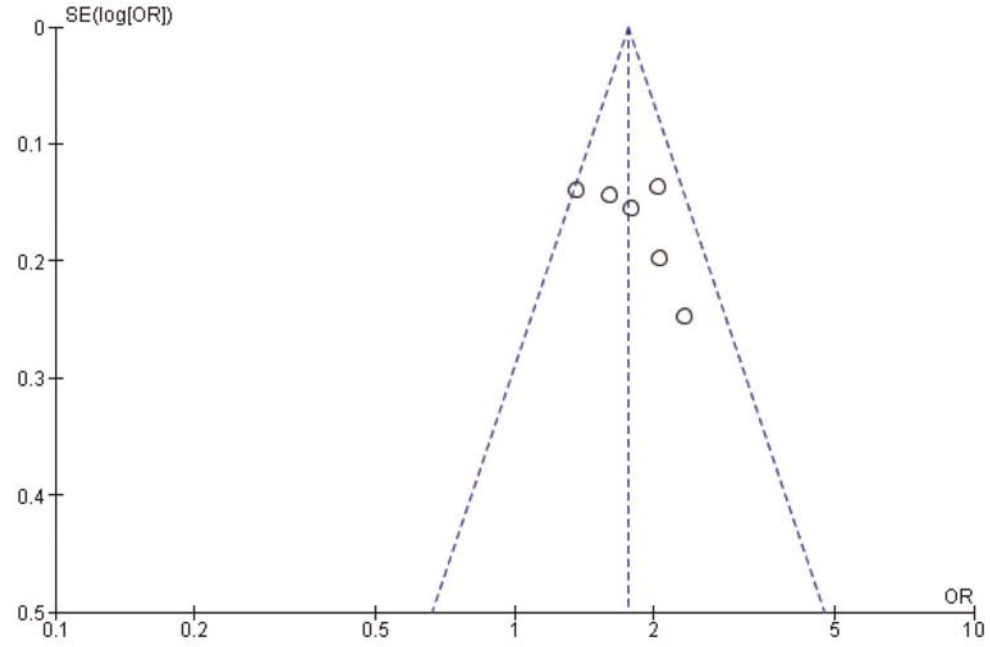
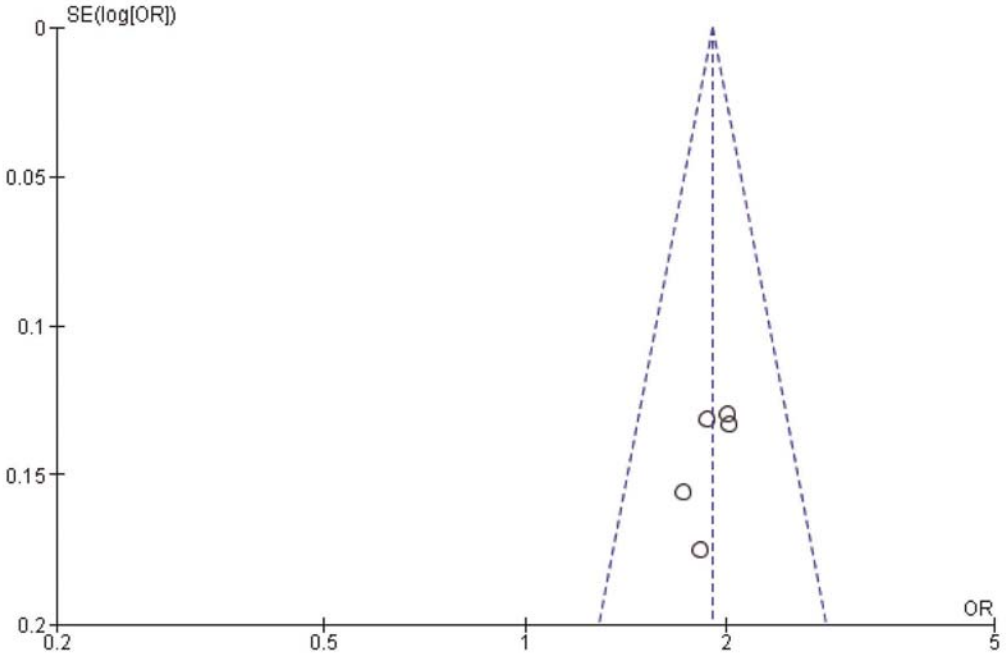
Supplementary material

Supplementary Fig. 1 Funnel plot for assessing bias for studies using AIAC for ADR.



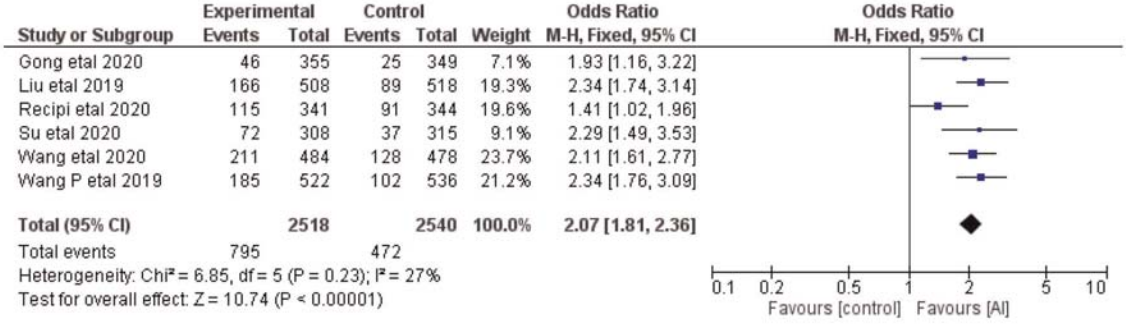
Supplementary Fig. 2 Funnel plot for assessing bias for studies using AIAC for PDR.



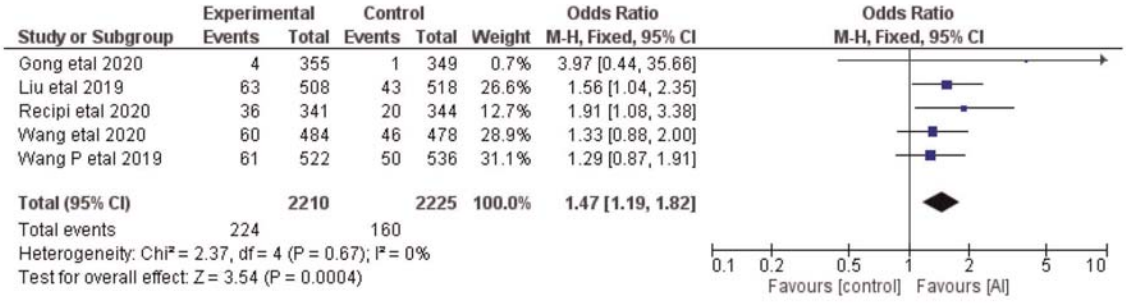
1

Supplementary material

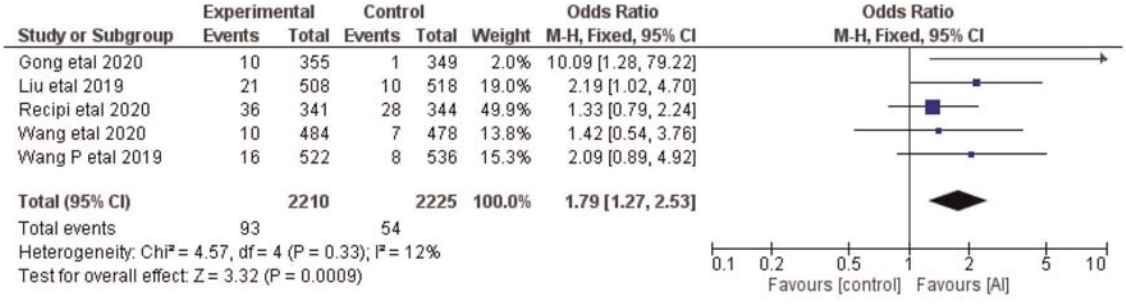
Supplementary Fig. 3 Forest plot for studies assessing the effect on artificial intelligence-aided colonoscopy compared to control (conventional colonoscopy) on adenoma detection rate for adenomas measuring **a** 0-5 mm in size, **b** 6-10 mm and **c** ≥ 10 mm. CI, confidence interval.



(A)



(B)



(C)

Supplementary material

Supplementary Table 1 Quality assessment of studies.

	Random Sequence Generation (Selection Bias)	Allocation Concealment (Selection Bias)	Blinding of Participants (Performance Bias)	Blinding of Outcome Assessment (Detection Bias)	Incomplete outcome Data (Attrition Bias)	Selective Reporting (Reporting Bias)
Wang Pu et al (2019)	+	-	-	+	+	+
Gong et al (2020)	+	-	+	+	+	+
Wang Pu et al (2020)	+	+	+	+	+	+
Liu et al (2019)	+	-	-	+	+	+
Su et al (2020)	+	-	+	+	+	+
Recipi et al (2020)	+	-	-	-	+	+

3

Supplementary Table 2 Quality of body of evidence: summary finding table (GRADE): Artificial intelligence-aided colonoscopy compared with conventional colonoscopy.

Certainty assessment							№ of patients		Effect		Certainty
No. of studies	**Study design**	**Risk of bias**	**Inconsistency**	**Indirectness**	**Imprecision**	**Other considerations**	**artificial intelligence aided colonoscopy**	**conventional colonoscopy**	**Relative (95% CI)**	**Absolute (95% CI)**	
6	randomized trials	serious [a]	not serious	not serious	not serious	none	850/2518 (33.8%)	583/2540 (23.0%)	OR 1.76 (1.55 to 2.00)	**114 more per 1,000** (from 86 more to 144 more)	⨁⨁⨁◯ MODERATE
								0.0%		**0 fewer per 1,000** (from 0 fewer to 0 fewer)	
5	randomized trials	serious [b]	not serious	not serious	not serious	none	933/2177 (42.9%)	674/2196 (30.7%)	OR 1.90 (1.68 to 2.15)	**150 more per 1,000** (from 120 more to 181 more)	⨁⨁⨁◯ MODERATE

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Supplementary material

Certainty assessment							№ of patients		Effect		Certainty
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	artificial intelligence aided colonoscopy	conventional colonoscopy	Relative (95% CI)	Absolute (95% CI)	
								0.0%		**0 fewer per 1,000** (from 0 fewer to 0 fewer)	
6	randomized trials	serious [b]	not serious	not serious	not serious	none	582/2518 (23.1%)	369/2540 (14.5%)	OR 1.81 (1.57 to 2.10)	**90 more per 1,000** (from 65 more to 118 more)	⨁⨁⨁◯ MODERATE
5	randomized trials	serious [b]	not serious	not serious	not serious	none	93/2210 (4.2%)	54/2225 (2.4%)	OR 1.79 (1.27 to 2.53)	**18 more per 1,000** (from 6 more to 35 more)	⨁⨁⨁◯ MODERATE
								0.0%		**0 fewer per 1,000** (from 0 fewer to 0 fewer)	

5

Supplementary material

Certainty assessment								№ of patients		Effect		Certainty
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	artificial intelligence aided colonoscopy	conventional colonoscopy	Relative (95% CI)	Absolute (95% CI)		
6	randomized trials	serious [a]	not serious	not serious	not serious	none	795/2518 (31.6%)	472/2540 (18.6%)	OR 2.07 (1.81 to 2.36)	**135 more per 1,000** (from 107 more to 164 more)	⊕⊕⊕◯ MODERATE	
								0.0%	not estimable	**0 fewer per 1,000** (from 0 fewer to 0 fewer)	-	
5	randomized trials	serious [a]	not serious	not serious	not serious	none	224/2210 (10.1%)	160/2225 (7.2%)	OR 1.47 (1.19 to 1.82)	**30 more per 1,000** (from 13 more to 52 more)	⊕⊕⊕◯ MODERATE	

Certainty assessment							№ of patients		Effect		Certainty
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	artificial intelligence aided colonoscopy	conventional colonoscopy	Relative (95% CI)	Absolute (95% CI)	
								0.0%		**0 fewer per 1,000** (from 0 fewer to 0 fewer)	

Explanations

a. This domain was downgraded by 1 point, because there was evidence of performance and detection bias in all studies included in this review

b. This domain was downgraded by 1 point, because there was evidence of performance and detection bias in all studies included in this review

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